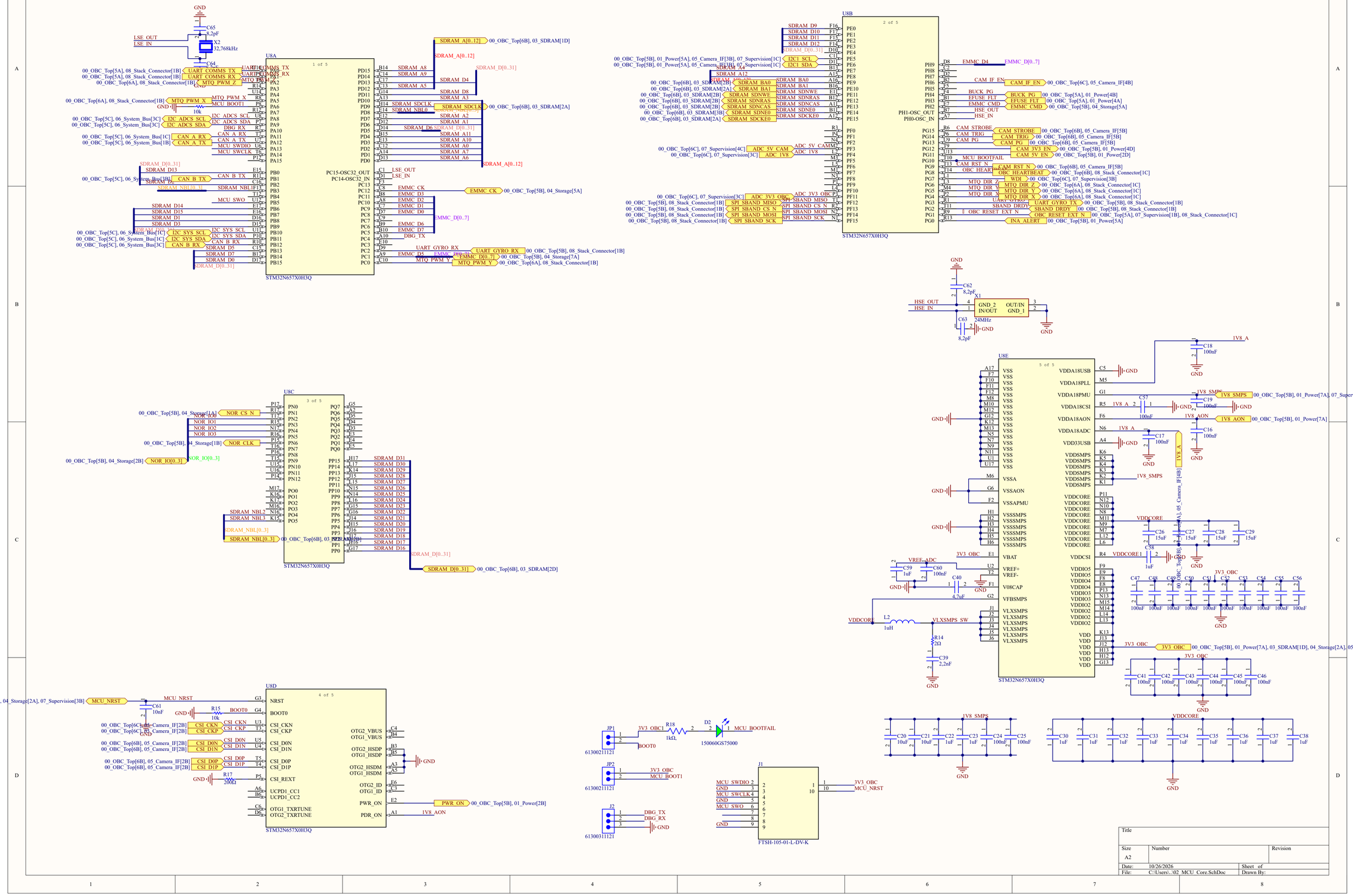

GND
C65
8.2pF
LSE_OUT
LSE_IN
X2
32.768kHz
C64
6pF
U8A
1 of 5
STM32N657X0H3Q
00_OBC_Top[5A], 08_Stack_Connector[1B] UART_COMMS_TX
00_OBC_Top[5A], 08_Stack_Connector[1B] UART_COMMS_RX
00_OBC_Top[6A], 08_Stack_Connector[1B] MTQ_PWM_Z
MTQ_PWM_X
MCU_BOOT1
10k
00_OBC_Top[5C], 06_System_Bus[3C] I2C_ADCS_SCL
00_OBC_Top[5C], 06_System_Bus[3C] I2C_ADCS_SDA
DBG_RX
CAN_A_RX
00_OBC_Top[5C], 06_System_Bus[1C] CAN_A_TX
00_OBC_Top[5C], 06_System_Bus[1B]
MCU_SWDIO
MCU_SWCLK
SDRAM_D[0..31]
SDRAM_D13
00_OBC_Top[5C], 06_System_Bus[1B] CAN_B_TX
SDRAM_D2
SDRAM_NBL[0..3]
MCU_SWO
SDRAM_D14
SDRAM_D15
SDRAM_D1
SDRAM_D3
SDRAM_D10
00_OBC_Top[5C], 06_System_Bus[1C] I2C_SYS_SCL
00_OBC_Top[5C], 06_System_Bus[1C] I2C_SYS_SDA
00_OBC_Top[5C], 06_System_Bus[3C] CAN_B_RX
SDRAM_D5
SDRAM_D7
SDRAM_D0
SDRAM_A[0..12]
00_OBC_Top[6B], 03_SDRAM[1D]
SDRAM_A[0..12]
SDRAM_D[0..31]
SDRAM_D4
SDRAM_D8
SDRAM_A3
SDRAM_SDCLK
00_OBC_Top[6B], 03_SDRAM[2A]
SDRAM_NBL0
SDRAM_A2
SDRAM_A1
SDRAM_D6
SDRAM_D[0..31]
SDRAM_A11
SDRAM_A10
SDRAM_A0
SDRAM_A7
SDRAM_A6
SDRAM_A[0..12]
PC15-OSC32_OUT
PC14-OSC32_IN
LSE_OUT
LSE_IN
EMMC_CK
00_OBC_Top[5B], 04_Storage[5A]
EMMC_D3
EMMC_D2
EMMC_D1
EMMC_D0
EMMC_D[0..7]
EMMC_D6
EMMC_D7
DBG_TX
UART_GYRO_RX
UART_GYRO_RX
00_OBC_Top[5B], 08_Stack_Connector[1B]
EMMC_D5
EMMC_D[0..7]
00_OBC_Top[5B], 04_Storage[7A]
MTQ_PWM_Y
MTQ_PWM_Y
00_OBC_Top[6A], 08_Stack_Connector[1B]
U8B
2 of 5
SDRAM_D9
SDRAM_D10
SDRAM_D11
SDRAM_D12
SDRAM_D[0..31]
00_OBC_Top[5B], 01_Power[5A], 05_Camera_IF[3B], 07_Supervision[1C] I2C1_SCL
00_OBC_Top[5B], 01_Power[5A], 05_Camera_IF[3B], 07_Supervision[1C] I2C1_SDA
SDRAM_A12
SDRAM_A[0..12]
00_OBC_Top[6B], 03_SDRAM[2B] SDRAM_BA0
00_OBC_Top[6B], 03_SDRAM[2A] SDRAM_BA1
00_OBC_Top[6B], 03_SDRAM[2B] SDRAM_SDNWE
00_OBC_Top[6B], 03_SDRAM[2B] SDRAM_SDNRAS
00_OBC_Top[6B], 03_SDRAM[2B] SDRAM_SDNCAS
00_OBC_Top[6B], 03_SDRAM[3B] SDRAM_SDNE0
00_OBC_Top[6B], 03_SDRAM[2A] SDRAM_SDCKE0
00_OBC_Top[6C], 07_Supervision[4C] ADC_5V_CAM
00_OBC_Top[6C], 07_Supervision[3C] ADC_1V8
00_OBC_Top[6C], 07_Supervision[3C] ADC_3V3_OBC
00_OBC_Top[5B], 08_Stack_Connector[1B] SPI_SBAND_MISO
00_OBC_Top[5B], 08_Stack_Connector[1B] SPI_SBAND_CS_N
00_OBC_Top[5B], 08_Stack_Connector[1B] SPI_SBAND_MOSI
00_OBC_Top[5B], 08_Stack_Connector[1B] SPI_SBAND_SCK
EMMC_D[0..7]
EMMC_D4
CAM_IF_EN
CAM_IF_EN
00_OBC_Top[6C], 05_Camera_IF[4B]
BUCK_PG
BUCK_PG
00_OBC_Top[5A], 01_Power[4B]
EFUSE_FLT
EFUSE_FLT
00_OBC_Top[5A], 01_Power[4A]
EMMC_CMD
EMMC_CMD
00_OBC_Top[5B], 04_Storage[5A]
HSE_OUT
HSE_IN
CAM_STROBE
CAM_STROBE
00_OBC_Top[6B], 05_Camera_IF[5B]
CAM_TRIG
CAM_TRIG
00_OBC_Top[6B], 05_Camera_IF[5B]
CAM_PG
CAM_PG
00_OBC_Top[6B], 05_Camera_IF[5B]
CAM_3V3_EN
00_OBC_Top[5B], 01_Power[4D]
CAM_5V_EN
00_OBC_Top[5B], 01_Power[2D]
MCU_BOOTFAIL
CAM_RST_N
CAM_RST_N
00_OBC_Top[6B], 05_Camera_IF[5B]
OBC_HEARTBEAT
OBC_HEARTBEAT
00_OBC_Top[6B], 08_Stack_Connector[1C]
MTQ_DIR_Z
MTQ_DIR_Z
00_OBC_Top[6A], 08_Stack_Connector[1C]
MTQ_DIR_Y
MTQ_DIR_Y
00_OBC_Top[6A], 08_Stack_Connector[1C]
MTQ_DIR_X
MTQ_DIR_X
00_OBC_Top[6A], 08_Stack_Connector[1C]
UART_GYRO_TX
00_OBC_Top[5B], 08_Stack_Connector[1B]
SBAND_DRDY
SBAND_DRDY
00_OBC_Top[5B], 08_Stack_Connector[1B]
OBC_RESET_EXT_N
OBC_RESET_EXT_N
00_OBC_Top[5A], 07_Supervision[1B], 08_Stack_Connector[1C]
INA_ALERT
00_OBC_Top[5B], 01_Power[5A]
STM32N657X0H3Q
U8C
3 of 5
00_OBC_Top[5B], 04_Storage[1A] NOR_CS_N
NOR_IO1
NOR_IO2
NOR_IO3
00_OBC_Top[5B], 04_Storage[1B] NOR_CLK
00_OBC_Top[5B], 04_Storage[2B] NOR_IO[0..3]
NOR_IO[0..3]
SDRAM_NBL2
SDRAM_NBL3
SDRAM_NBL[0..3]
00_OBC_Top[6B], 03_SDRAM[2B] SDRAM_NBL[0..3]
SDRAM_D31
SDRAM_D30
SDRAM_D29
SDRAM_D28
SDRAM_D27
SDRAM_D26
SDRAM_D25
SDRAM_D24
SDRAM_D23
SDRAM_D22
SDRAM_D21
SDRAM_D20
SDRAM_D19
SDRAM_D18
SDRAM_D17
SDRAM_D16
SDRAM_D[0..31]
SDRAM_D[0..31]
00_OBC_Top[6B], 03_SDRAM[2D]
STM32N657X0H3Q
U8D
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00_OBC_Top[5B], 04_Storage[2A], 07_Supervision[3B] MCU_NRST
MCU_NRST
BOOT0
00_OBC_Top[6C], 05_Camera_IF[2B] CSI_CKN
00_OBC_Top[6C], 05_Camera_IF[2B] CSI_CKP
00_OBC_Top[6B], 05_Camera_IF[2B] CSI_D0N
00_OBC_Top[6B], 05_Camera_IF[2B] CSI_D1N
00_OBC_Top[6B], 05_Camera_IF[2B] CSI_D0P
00_OBC_Top[6B], 05_Camera_IF[2B] CSI_D1P
CSI_REXT
UCPD1_CC1
UCPD1_CC2
OTG1_TXRTUNE
OTG2_TXRTUNE
OTG2_VBUS
OTG1_VBUS
OTG2_HSDP
OTG1_HSDP
OTG2_HSDM
OTG1_HSDM
OTG2_ID
OTG1_ID
PWR_ON
PDR_ON
PWR_ON
00_OBC_Top[5B], 01_Power[2B]
1V8_AON
STM32N657X0H3Q
JP1
61300211121
3V3_OBC1
R18
1kΩ
D2
150060GS75000
MCU_BOOTFAIL
JP2
61300211121
3V3_OBC
MCU_BOOT1
J2
61300311121
DBG_TX
DBG_RX
J1
FTSH-105-01-L-DV-K
MCU_SWDIO
MCU_SWCLK
MCU_SWO
3V3_OBC
MCU_NRST
U8E
5 of 5
VDDA18USB
VDDA18PLL
VDDA18PMU
VDDA18CSI
VDDA18AON
VDDA18ADC
VDD33USB
VDDSMPS
VSS
VSSA
VSSAON
VSSAPMU
VSSSMPS
VBAT
VREF+
VREF-
V08CAP
VFBSMPS
VLXSMPS
VDDCORE
VDDIO5
VDDIO4
VDDIO3
VDDIO2
VDD
1V8_A
1V8_SMPS
1V8_AON
VDDCORE
00_OBC_Top[5B], 01_Power[7A], 07_Supervision[3C]
00_OBC_Top[5B], 01_Power[7A]
00_OBC_Top[5B], 01_Power[4A], 05_Camera_IF[4B]
VREF_ADC
VDDCORE
VLXSMPS_SW
L2
1uH
R14
2Ω
C39
2.2nF
STM32N657X0H3Q
C62
8.2pF
X1
GND_2
IN/OUT
OUT/IN
GND_1
C63
24MHz
8.2pF
C18
100nF
C19
100nF
C16
100nF
C17
100nF
C57
100nF
C58
1uF
C59
1uF
C60
100nF
C40
4.7uF
C26
15uF
C27
15uF
C28
15uF
C29
15uF
C47
C48
C49
C50
C51
C52
C53
C54
C55
C56
100nF
C41
C42
C43
C44
C45
C46
C20
10uF
C21
10uF
C22
1uF
C23
1uF
C24
100nF
C25
100nF
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1uF
C31
1uF
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C37
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C38
1uF
C61
10nF
R15
10k
R17
200Ω
Title
Size
A2
Number
Revision
Date:
10/26/2026
Sheet of
File:
C:\Users\...\02_MCU_Core.SchDoc
Drawn By: